

gain in real time. This is discussed in Section 7.3, which includes a presentation of α - β and α - β - γ filtering.

When the dynamics of the system are not perfectly known, then the Kalman filter may not provide acceptable state estimates. This can be addressed by giving more weight to recent measurements when updating the state estimate, and discounting measurements that arrived a long time ago. This is called the fading-memory filter and is discussed in Section 7.4. Finally, there may be other information about the states other than the system model. For example, there may be state constraints that we know must be satisfied. Section 7.5 discusses several ways to incorporate state equality constraints and state inequality constraints into the formulation of the Kalman filter.

7.1 CORRELATED PROCESS AND MEASUREMENT NOISE

Our derivation of the Kalman filter in Chapter 5 assumed that the process noise and measurement noise were uncorrelated. In this section, we will show how correlated process and measurement noise changes the Kalman filter equations. Suppose that we have a system given by

$$\begin{aligned}
 x_k &= F_{k-1}x_{k-1} + G_{k-1}u_{k-1} + w_{k-1} \\
 y_k &= H_k x_k + v_k \\
 w_k &\sim (0, Q_k) \\
 v_k &\sim (0, R_k) \\
 E[w_k w_j^T] &= Q_k \delta_{k-j} \\
 E[v_k v_j^T] &= R_k \delta_{k-j} \\
 E[w_k v_j^T] &= M_k \delta_{k-j+1}
 \end{aligned} \tag{7.1}$$

We see that the process noise in the system equation is correlated with the measurement noise, with the cross covariance given by $M_k \delta_{k-j+1}$. Our derivation in Chapter 5 assumed that M_k was zero, but in this section we will relax that assumption. For example, suppose that our system is an airplane and winds are buffeting the plane. We are using an anemometer to measure wind speed as an input to our Kalman filter. So the random gusts of wind affect both the process (i.e., the airplane dynamics) and the measurement (i.e., the sensed wind speed). We see that there is a correlation between the process noise and the measurement noise. From the above equation, we see that the process noise at time k is correlated with the measurement noise at time $(k+1)$; that is, w_k is correlated with v_{k+1} . This is because w_k affects the state at time $(k+1)$, just as v_{k+1} affects the measurement at time $(k+1)$.

In order to find the Kalman filter equations for the correlated noise system, we will define the estimation errors as

$$\begin{aligned}
 \epsilon_k^- &= x_k - \hat{x}_k^- \\
 \epsilon_k^+ &= x_k - \hat{x}_k^+
 \end{aligned} \tag{7.2}$$

As in our original Kalman filter derivation of Chapter 5, we still assume that our update equations for the state estimate are given as follows:

$$\begin{aligned}\hat{x}_k^- &= F_{k-1}\hat{x}_{k-1}^+ + G_{k-1}u_{k-1} \\ \hat{x}_k^+ &= \hat{x}_k^- + K_k(y_k - H_k\hat{x}_k^-)\end{aligned}\quad (7.3)$$

The gain matrix K_k will not be the same as we derived in Chapter 5, but the form of the measurement update equation is still the same. Equation (7.2) can be expanded using the above equations as

$$\begin{aligned}\epsilon_k^- &= x_k - \hat{x}_k^- \\ &= (F_{k-1}x_{k-1} + G_{k-1}u_{k-1} + w_{k-1}) - (F_{k-1}\hat{x}_{k-1}^+ + G_{k-1}u_{k-1}) \\ &= F_{k-1}\epsilon_{k-1}^+ + w_{k-1} \\ \epsilon_k^+ &= x_k - [\hat{x}_k^- + K_k(y_k - H_k\hat{x}_k^-)] \\ &= \epsilon_k^- - K_k(H_kx_k + v_k - H_k\hat{x}_k^-) \\ &= \epsilon_k^- - K_k(H_k\epsilon_k^- + v_k)\end{aligned}\quad (7.4)$$

The *a priori* and *a posteriori* estimation-error covariances can be written as

$$\begin{aligned}P_k^- &= E[\epsilon_k^-(\epsilon_k^-)^T] \\ &= F_{k-1}P_{k-1}^+F_{k-1}^T + Q_{k-1} \\ P_k^+ &= E[\epsilon_k^+(\epsilon_k^+)^T] \\ &= E\{[\epsilon_k^- - K_k(H_k\epsilon_k^- + v_k)][\epsilon_k^- - K_k(H_k\epsilon_k^- + v_k)]^T\} \\ &= P_k^- - K_kH_kP_k^- - K_kE[v_k(\epsilon_k^-)^T] - P_k^-H_k^TK_k^T + \\ &\quad K_kH_kP_k^-H_k^TK_k^T + K_kE[v_k(\epsilon_k^-)^T]H_k^TK_k^T - \\ &\quad E(\epsilon_k^-v_k^T)K_k^T + K_kH_kE(\epsilon_k^-v_k^T)K_k^T + K_kE(v_kv_k^T)K_k^T\end{aligned}\quad (7.5)$$

In order to simplify this expression for P_k^+ , we need to find an expression for $E(\epsilon_k^-v_k^T)$. This can be computed as

$$\begin{aligned}E(\epsilon_k^-v_k^T) &= E[(x_k - \hat{x}_k^-)v_k^T] \\ &= E(x_kv_k^T - \hat{x}_k^-v_k^T) \\ &= E[(F_{k-1}x_{k-1} + G_{k-1}u_{k-1} + w_{k-1})v_k^T] - E[\hat{x}_k^-v_k^T] \\ &= 0 + 0 + M_k - 0\end{aligned}\quad (7.6)$$

In the above equation, the first term is 0 because x_{k-1} is independent of v_k , and v_k is zero-mean. The second term is 0 because u_{k-1} is independent of v_k . The last term is 0 because the *a priori* state estimate at time k is independent of v_k . Substituting this expression for $E(\epsilon_k^-v_k^T)$ into Equation (7.5) gives

$$\begin{aligned}P_k^+ &= P_k^- - K_kH_kP_k^- - K_kM_k^T - P_k^-H_k^TK_k^T + K_kH_kP_k^-H_k^TK_k^T + \\ &\quad K_kM_k^TH_k^TK_k^T - M_kK_k^T + K_kH_kM_kK_k^T + K_kR_kK_k^T \\ &= (I - K_kH_k)P_k^-(I - K_kH_k)^T + K_kR_kK_k^T + \\ &\quad K_k(H_kM_k + M_k^TH_k^T)K_k^T - M_kK_k^T - K_kM_k^T\end{aligned}\quad (7.7)$$

Now we need to find the gain matrix K_k that minimizes $\text{Tr}(P_k^+)$. Recall from Equation (1.66) that

$$\frac{\partial \text{Tr}(ABA^T)}{\partial A} = 2AB \text{ if } B \text{ is symmetric} \quad (7.8)$$

We can use this fact to derive

$$\begin{aligned}
\frac{\partial \text{Tr}(P_k^+)}{\partial K_k} &= -2(I - K_k H_k) P_k^- H_k^T + 2K_k R_k + \\
&\quad 2K_k (H_k M_k + M_k^T H_k^T) - M_k - M_k^T \\
&= 2 [K_k (H_k P_k^- H_k^T + H_k M_k + M_k^T H_k^T + R_k) - \\
&\quad P_k^- H_k^T - M_k] \quad (7.9)
\end{aligned}$$

In order to make this partial derivative zero, we need to set the gain K_k as follows:

$$K_k = (P_k^- H_k^T + M_k)(H_k P_k^- H_k^T + H_k M_k + M_k^T H_k^T + R_k)^{-1} \quad (7.10)$$

This gives the optimal Kalman gain matrix for the system with correlated process and measurement noise. The estimation-error covariance is then obtained from Equation (7.7) as

$$\begin{aligned}
P_k^+ &= (I - K_k H_k) P_k^- (I - K_k H_k)^T + \\
&\quad K_k (H_k M_k + M_k^T H_k^T + R_k) K_k^T - M_k K_k^T - K_k M_k^T \\
&= P_k^- - K_k H_k P_k^- - P_k^- H_k^T K_k^T + \\
&\quad K_k (H_k P_k^- H_k^T + H_k M_k + M_k^T H_k^T + R_k) K_k^T - \\
&\quad M_k K_k^T - K_k M_k^T \\
&= P_k^- - K_k (H_k P_k^- + M_k^T) - (P_k^- H_k^T + M_k) K_k^T + \\
&\quad (P_k^- H_k^T + M_k)(H_k P_k^- H_k^T + H_k M_k + M_k^T H_k^T + R_k)^{-1} (H_k P_k^- + M_k^T) \\
&= P_k^- - K_k (H_k P_k^- + M_k^T) - (P_k^- H_k^T + M_k) K_k^T + (P_k^- H_k^T + M_k) K_k^T \\
&= P_k^- - K_k (H_k P_k^- + M_k^T) \quad (7.11)
\end{aligned}$$

This gives the measurement-update equation for the estimation-error covariance for the Kalman filter with correlated process and measurement noise. The measurement-update equation for the state estimate is the same as for the standard Kalman filter and is given in Equation (7.3). The time-update equations for the state estimate and the estimation-error covariance are also the same as before. The Kalman filter for the system with correlated process and measurement noise can be summarized as follows.

The general discrete-time Kalman filter

1. The system and measurement equations are given as

$$\begin{aligned}
x_k &= F_{k-1} x_{k-1} + G_{k-1} u_{k-1} + w_{k-1} \\
y_k &= H_k x_k + v_k \\
w_k &\sim (0, Q_k) \\
v_k &\sim (0, R_k) \\
E[w_k w_j^T] &= Q_k \delta_{k-j} \\
E[v_k v_j^T] &= R_k \delta_{k-j} \\
E[w_k v_j^T] &= M_k \delta_{k-j+1} \quad (7.12)
\end{aligned}$$

2. The Kalman filter is initialized as

$$\begin{aligned}\hat{x}_0^+ &= E(x_0) \\ P_0^+ &= E[(x_0 - \hat{x}_0^+)(x_0 - \hat{x}_0^+)^T]\end{aligned}\quad (7.13)$$

3. For each time step $k = 1, 2, \dots$, the Kalman filter equations are given as

$$\begin{aligned}P_k^- &= F_{k-1}P_{k-1}^+F_{k-1}^T + Q_{k-1} \\ K_k^+ &= (P_k^-H_k^T + M_k)(H_kP_k^-H_k^T + H_kM_k + M_k^TH_k^T + R_k)^{-1} \\ &= P_k^+(H_k^T + (P_k^-)^{-1}M_k)(R_k - M_k^T(P_k^-)^{-1}M_k)^{-1} \\ \hat{x}_k^- &= F_{k-1}\hat{x}_{k-1}^+ + G_{k-1}u_{k-1} \\ \hat{x}_k^+ &= \hat{x}_k^- + K_k(y_k - H_k\hat{x}_k^-) \\ P_k^+ &= (I - K_kH_k)P_k^-(I - K_kH_k)^T + \\ &\quad K_k(H_kM_k + M_k^TH_k^T + R_k)K_k^T - M_kK_k^T - K_kM_k^T \\ &= \left[(P_k^-)^{-1} + (H_k^T + (P_k^-)^{-1}M_k)(R_k - M_k^T(P_k^-)^{-1}M_k)^{-1} \times \right. \\ &\quad \left. (H_k + M_k^T(P_k^-)^{-1}) \right]^{-1} \\ &= P_k^- - K_k(H_kP_k^- + M_k^T)\end{aligned}\quad (7.14)$$

The second form for P_k^+ and the second form for K_k can be derived by following a procedure similar to that shown in Section 3.3.1. Note that this is a generalization of the Kalman filter that was presented in Equation (5.19). If $M_k = 0$, then the above equations reduce to Equation (5.19).

■ EXAMPLE 7.1

Consider the following scalar system:

$$\begin{aligned}x_k &= 0.8x_{k-1} + w_{k-1} \\ y_k &= x_k + v_k \\ E[w_k w_j^T] &= 1\delta_{k-j} \\ E[v_k v_j^T] &= 0.1\delta_{k-j} \\ E[w_k v_j^T] &= M\delta_{k-j+1}\end{aligned}\quad (7.15)$$

We can use the method discussed in Section 2.7 to simulate correlated noise. The Kalman filter equations given above can then be run to obtain an estimate of the state. Table 7.1 shows (for several values of M) the variance of the estimation error for the standard Kalman filter (when $M = 0$ is assumed) and for the correlated noise Kalman filter (when the correct value of M is used). When $M = 0$, the estimation-error variances are the same for the two filters, as expected. However, when $M \neq 0$, the filter that uses the correct value of M performs noticeably better than the filter that incorrectly assumes that $M = 0$.

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